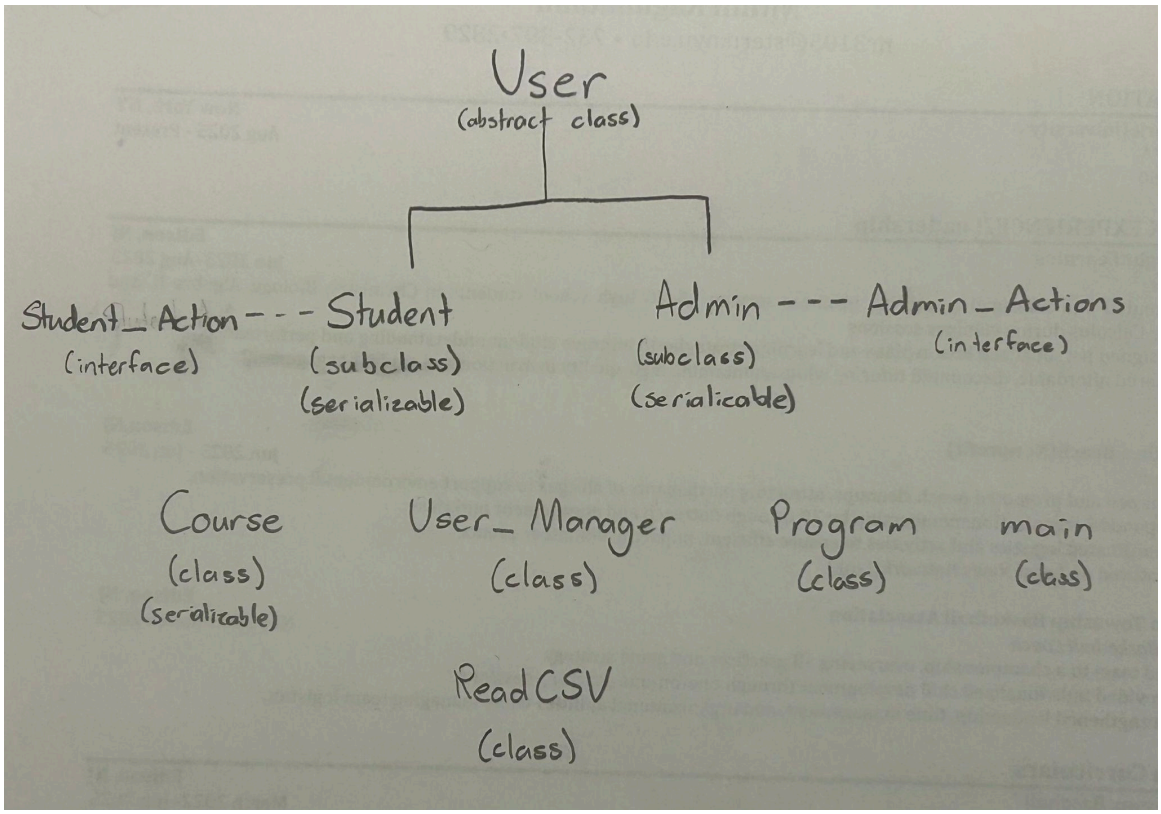
# Technical Design Document

- **main.java →**
  - runs the static method Program.program.
- **Program.java →**
  - has a single method that runs the following: starts by loading data from CSV or deserializing serialized files for objects, which is stored in the User_Manager. Then, enter into a menu that allows login, account creation, and closing the program. logging in calls the .menu method in either the admin or student class. Closing the program serializes all objects created in session.
- **ReadCSV.java →**
  - contains a single method that reads from a CSV using a scanner and returns an ArrayList of courses.
- **User_Manager.java →**
  - contains the ArrayList of all students and an admin object as instance variables. has methods to add a new student, add an admin, and login for either type of user (student or admin). Each method works via command line interface input (CLI).
- **User.java (serializable) →**
  - abstract class that contains the declarations of all fields and methods that the admin and student use. has an overloaded constructor that allows for object creation with all normal fields (username, password, first_name, last_name) or with an added attribute (birthday). method stub is for the menu method that gets implemented by both subclasses of User.
- **Student.java (serializable) →**
  - subclass of User. implements interface Student_Action. has two constructors for w/ and w/o birthdays. has methods for students to view all courses, view non-full courses, register for a course, withdraw from a course, view all courses they're registered for, and

finally a menu method. The menu method enters the user into a while loop that allows them to select any of the previously mentioned methods to perform OR exit the menu.

- **Admin.java (serializable) →**
  - subclass of User. implements interface Admin_Actions. has methods for admin to view courses; search for, add, delete, edit, display info for, register a student for a course; view all full courses; write all courses to a CSV file; view course roster; view all courses a student has registered for; and a menu. The menu method has the main functionality for the admin. The menu is split into two submenus: course management and reports. course management allows for adding, deleting, editing, displaying info for, registering a certain student for a course; registering a new student; or exiting the submenu. reports allows for viewing all courses, viewing all courses that are full, viewing the names of the students that are registered in a specific course, viewing the list of courses a specific student is registered for, sorting courses based on the current number of student registers, and exiting the menu.
- **Course.java (serializable) →**
  - has instance variables for course name, ID, max students, # of students, an ArrayList<Student> of students, an ArrayList<String> of student names, instructor name, section number, and class location. has a constructor that sets all fields to those that are set during Course object instantiation.
- **Admin_Actions.java (interface) →**
  - interface that provides method stubs for all operations that the Admin class implements.
- **Student_Action.java (interface) →**
  - interface that provides method stubs for all operations that the Student class implements.

# UML Diagram


User
(abstract class)
Student_Action
(interface)
Student
(subclass)
(serializable)
Admin
(subclass)
(serializable)
Admin_Actions
(interface)
Course
(class)
(serializable)
User_Manager
(class)
Program
(class)
main
(class)
ReadCSV
(class)